THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET B

Chapter 11: **Keeping Time with the Skies** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The Moon completes its cycle of phases in about: 1

(a) 354 days **(b)** 29.5 days **(c)** 7 days **(d)** 365 days

2. The phase when the whole lit face is visible is the: 1

(a) crescent **(b)** new moon **(c)** half moon **(d)** full moon

3. The Moon rises about ___ later each day. 1

(a) 5 minutes **(b)** 5 hours **(c)** 12 hours **(d)** 50 minutes

4. Earth's revolution around the Sun takes about: 1

(a) 365¼ days **(b)** 354 days **(c)** 24 hours **(d)** 29.5 days

5. The extra month added in some calendars is called: 1

(a) Solstice **(b)** Leap day **(c)** Adhika Maasa **(d)** Equinox

6. Which of these is NOT a leap year? 1

(a) 1900 **(b)** 1600 **(c)** 2024 **(d)** 2000

7. The sidereal year is longer than the tropical year by about: 1

(a) 20 days **(b)** 20 minutes **(c)** 20 hours **(d)** 2 days

8. Festivals like Makar Sankranti and Pongal follow a: 1

(a) solar calendar **(b)** random calendar
(c) lunar calendar only **(d)** no calendar

9. An artificial satellite is: 1

(a) a comet **(b)** a human-made object orbiting Earth
(c) a star **(d)** a natural moon

10. The 'Father of the Indian Space programme' is: 1

(a) Homi Bhabha **(b)** S. Chandrasekhar **(c)** Vikram Sarabhai **(d)** C.V. Raman

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is a leap year? Why is it needed? 2

12. Why is a lunar year shorter than a solar year? 2

13. What is Adhika Maasa? 2

14. When does the Indian National Calendar year begin? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. Compare the lengths of a lunar year, solar year and the gap. How is the gap corrected? 3

16. Describe the Indian National Calendar. 3

17. Why do festivals like Eid-ul-fitr occur on different Gregorian dates each year? 3

OR

Explain how the Sun's rising point shows the seasons. 3

18. What are artificial satellites? Give three uses. 3

OR

Name three Indian satellites and their purpose. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

19. Explain how observations of the Sun and stars helped ancient people create calendars. 5

OR

Explain the difference between tropical and sidereal years and why it matters for calendars. 5

20. Why do some festivals fall on nearly the same Gregorian date each year while others move? Give examples. 5

OR

Describe artificial satellites: how they are launched, their types by use, and examples from India. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND